

Agent Confidence Entropy as a Pre-hoc Difficulty Oracle for Multi-Agent Group Generation: Bayesian Agentic Hyperparameter Tuning on Perturb-Seq Experimental Design

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Abstract

We study test-time compute (TTC) scaling in the multi-agent setting, taking as our testbed a five-agent orchestrator for perturb-seq experimental design inspired by CELLFORGE [Chen et al., 2025] and operated via the MASSGEN framework [Li and contributors, 2025]. We introduce four lightweight, compute-free metrics derived from the agents’ own per-round confidence-and-critique trace — Agent Confidence Entropy (ACE), Critique Severity Dispersion (CSD), a convergence signal based on the round-over-round change in mean confidence (ΔC), and a Winner Flip Rate (WFR) — and combine them into a composite Task Difficulty Index (TDI). Every reported number in this paper comes from real Perturb-seq data: all three ADAMSON 2016 [Adamson et al., 2016] scPerturb subsets (97 TF perturbations stratified by mean $|\Delta \log_{10} p|$ on the target gene into 21 held-out tasks) and NORMAN 2019 [Norman et al., 2019] (15 singletons + 5 doublets stratified by guide-count quartile, seed = 2026). We run 1944 trainer cells ($\{\text{LINEAR, MLP, SCGPT_SMALL}\} \times N \in \{3, 5\} \times R \in \{1, 2, 3\} \times 3$ seeds per task) plus 108 end-to-end agentic lifecycle runs in which five free-acting LLM agents drawn from a free-tier OpenRouter rotation pool (Nemotron-3 Super 120B, Ling-2.6 1T MoE, Hermes-3 405B, GPT-OSS 120B, Qwen3-Next 80B, Llama 3.3 70B, Gemma-4 31B / 3-27B) make genuine per-task choices over backbone, HVG count, learning rate, ridge λ and epochs, with the Validator returning a structured critique that the next round’s Architect applies as a configuration delta. Two of three pre-registered gates pass: median best-config MSD on the top-20 DEGs is $0.147 < 0.20$ on Adamson and $0.131 < 0.30$ on Norman. Architect backbone-choice entropy is 0.36 nats — agents converge on SCGPT_SMALL for 91% of unseen K562 perturbations while exercising independent variation on the data-curation axis (HVG-count entropy 0.30 nats). We expressly do not generate any synthetic Perturb-seq cells: every record consumed by every metric is sourced from the published scPerturb Zenodo bundle (record 13350497) via SHA-gated fetchers. Total Modal A100 cost: \$4.04 under a \$28 hard kill; LLM cost: \$0. Code, frozen datasets, and reproducibility artifacts are released with the paper.

1 Introduction

Test-time compute (TTC) has emerged as a first-class axis of capability in large language models. Snell et al. [2024] showed that an *optimally allocated* per-prompt TTC budget lets a small model match a $14\times$ larger model under FLOPs-matched comparison, with the optimal allocation depending on the difficulty of the prompt. The recent taxonomy of Zhang et al. [2025] organises the exploding TTC literature along four orthogonal axes — *what* to scale, *how* to scale, *where* to scale, and *how well* the method scales — but the prominent applications to date address *single-chain* reasoning: parallel samples scored by a process reward model (PRM) [Lightman et al., 2024], sequential self-revision, tree search [Yao et al., 2023], or internal chain-of-thought trained via reinforcement learning [DeepSeek-AI et al., 2025, Muennighoff et al., 2025].

The multi-agent case is structurally different but no less TTC-laden. A group-generation system such as CELLFORGE [Chen et al., 2025] dispatches several specialised agents — typically a data curator, a literature/prior agent, a model architect, a trainer, and a biological validator — and coordinates them through propose-critique-vote rounds. Every such system today spends compute *uniformly*: the same

number of agents for the same number of rounds regardless of whether the task is GSK3B knockout (well-studied, linear pathway) or a novel combinatorial perturbation. *This paper asks whether the group’s own internal dynamics carry enough signal to allocate its compute adaptively.*

Contributions. We make four contributions, all within the empirical frame of perturb-seq experimental design [Norman et al., 2019, Adamson et al., 2016] but generalising to any propose-critique-vote system:

1. We formalise four easy-to-compute observables of a multi-agent run — ACE, CSD, $\Delta C/\Delta ACE$, and WFR — and combine them into a scalar Task Difficulty Index (TDI).
2. We introduce a *preflight probe*: a single shallow round whose four-dimensional signature is intended to serve as a sufficient statistic for a Bayesian recommender over orchestration hyperparameters $\varphi = (N, R, \text{backbone})$.
3. We evaluate every claim end-to-end on *real* Perturb-seq data — ADAMSON 2016 (all three scPerturb subsets) and NORMAN 2019 singletons + doublets — over 1944 trainer cells and 108 full agentic lifecycle runs on a single Modal A100, within a \$28 hard-kill budget. We report both the gates that pass and the one that fails honestly.
4. We release the calibration harness, the MASSGEN-installable evaluation skill that produces metric reports on any propose-critique run, and the SHA-gated dataset fetchers + atomic JSONL trace logs that make the entire run reproducible from a single Modal command.

Scope. Every empirical claim in this paper is anchored to real ADAMSON 2016 [Adamson et al., 2016] and NORMAN 2019 [Norman et al., 2019] K562 Perturb-seq cells sourced from the published scPerturb [Peidli et al., 2024] Zenodo bundle (record 13350497). *No synthetic Perturb-seq data is generated or consumed by any reported result.* Genome-scale validation on REPLOGLE 2022 [Replogle et al., 2022] is the natural next step and is enumerated in §10; the paper deliberately does not extrapolate beyond K562.

2 Background and Related Work

Test-time compute scaling. The modern TTC literature is usefully organised by whether the extra compute is spent *in parallel* (best-of- N with a verifier [Cobbe et al., 2021, Lightman et al., 2024], self-consistency voting [Wang et al., 2023]), *sequentially* (iterative revision, self-refine), *hybrid* (tree search, MCTS, [Yao et al., 2023]), or *internal* to the model ([DeepSeek-AI et al., 2025, Muennighoff et al., 2025]); see Zhang et al. [2025] for a taxonomy and Wu et al. [2024], Liu et al. [2025] for scaling-law analyses. Snell et al. [2024] is the primary reference for the difficulty-conditional compute-optimal idea that our work extends to the group setting.

Multi-agent TTC. AUTOGEN [Wu et al., 2023] and MASSGEN [Li and contributors, 2025] generalise propose-critique-vote to heterogeneous agent pools; ARCHON [Saad-Falcon et al., 2024] and MAV [Lifshitz et al., 2025] treat inference-time architecture *itself* as the search space. CELLFORGE [Chen et al., 2025] applies this pattern specifically to single-cell experimental design. None of these works, to our knowledge, use the group’s internal confidence/critique distribution as an explicit difficulty signal.

Perturb-seq and single-cell foundation models. Perturb-seq couples CRISPR perturbations with single-cell RNA-sequencing readouts, offering a clean testbed for agentic scientific reasoning because ground-truth hardness can be proxied from literature density, replicate agreement, and donor-split transfer [Norman et al., 2019, Adamson et al., 2016, Replogle et al., 2022]. Backbones such as scGPT [Cui et al., 2024], GENEFORMER [Theodoris et al., 2023], scFOUNDATION [Hao et al., 2024], and scPRINT [Kalfon et al., 2025] provide pre-trained representations that the architect agent can configure.

3 Problem Setup

We consider an orchestrator with N specialised agents $\{\mathcal{A}_i\}_{i=1}^N$ coordinated over R rounds. At each round r , every agent produces a structured proposal with scalar confidence $c_i(r) \in [0, 1]$ and every other agent emits a critique with severity $S_{ij}(r) \in [0, 1]$ (row i critic, column j target; self-critique is undefined). A

winner index $w(r)$ is assigned by the orchestrator (typically $\operatorname{argmax} c - \bar{S}$, where $\bar{S}$ is the mean severity received). We let $\mathbf{C}(r) \in \mathbb{R}^N$ and $\mathbf{S}(r) \in \mathbb{R}^{N \times (N-1)}$ denote the per-round tuples.

A task t has a latent intrinsic difficulty $d_t \in [0, 1]$. The team’s downstream utility on the task is proxied by held-out validation AUROC $u(t, \varphi)$, where $\varphi = (N, R, b)$ with b a backbone choice. *Agentic hyperparameter tuning* is the problem of picking $\varphi^* \in \Phi$ minimising compute subject to a target utility — the multi-agent analogue of Snell et al. [2024]’s compute-optimal single-chain allocation.

4 Metrics

4.1 Agent Confidence Entropy (ACE)

Let $p(r) = \operatorname{softmax}(\mathbf{C}(r)/\tau)$ be the softmax of the confidence vector at round r with temperature τ (we use $\tau = 1$ throughout). ACE is the Shannon entropy of this distribution, normalised to $[0, 1]$ by its maximum:

$$\operatorname{ACE}(r) = - \sum_{i=1}^N p_i(r) \log p_i(r), \quad \operatorname{ACE}_{\text{norm}}(r) = \operatorname{ACE}(r) / \log N. \quad (1)$$

$\operatorname{ACE}_{\text{norm}} \rightarrow 0$ when one agent dominates (a crisp hierarchy among agents) and $\rightarrow 1$ when the distribution is flat. Crucially, a flat distribution has two very different causes — *confident consensus* (easy task) and *uniform uncertainty* (hard task) — so ACE must be read alongside the mean confidence.

4.2 Critique Severity Dispersion (CSD)

Let $\mathbf{S}(r)$ collect the off-diagonal entries of the critique matrix. We define CSD as the population variance of its entries:

$$\operatorname{CSD}(r) = \operatorname{Var}(\mathbf{S}(r)). \quad (2)$$

Large CSD concentrates disagreement: a small number of severe critiques weighs more than uniformly moderate ones. We also track the row- and column-variances for diagnostic use (which critic is inconsistent? which proposal is contested?).

4.3 Convergence signals

For a run with R rounds we track three convergence signals:

$$\Delta \operatorname{ACE} \triangleq \operatorname{ACE}_{\text{norm}}(R) - \operatorname{ACE}_{\text{norm}}(0), \quad (3)$$

$$\Delta C \triangleq \bar{\mathbf{C}}(R) - \bar{\mathbf{C}}(0), \quad (4)$$

$$\operatorname{WFR} \triangleq \frac{1}{R-1} \sum_{r=2}^R \mathbb{1}[w(r) \neq w(r-1)]. \quad (5)$$

A well-posed run has $\Delta \operatorname{ACE} < 0$, $\Delta C > 0$, and $\operatorname{WFR} \approx 0$. We use $1 - \min(\max(\Delta C, 0), 1)$ as a bounded “lack of convergence” feature.

4.4 Task Difficulty Index

We combine the four signals linearly:

$$\operatorname{TDI} = \alpha \operatorname{ACE}_{\text{norm}}(R) + \beta \operatorname{CSD}(R) + \gamma (1 - \min(\max(\Delta C, 0), 1)) + \delta \operatorname{WFR}, \quad (6)$$

with non-negative weights that sum to unity so $\operatorname{TDI} \in [0, 1]$. We compare two weight regimes: the hand-set *default* $(\alpha, \beta, \gamma, \delta) = (0.35, 0.25, 0.25, 0.15)$ and the *calibrated* weights fitted by ridge regression of (6) against a ground-truth difficulty label.

5 Preflight Probe and Bayesian Recommender

Probe signature. Given a run, the round-0 probe signature is $\mathbf{x}_0 = (\operatorname{ACE}_{\text{norm}}(0), \bar{\mathbf{C}}(0), \max_i c_i(0), \operatorname{CSD}(0)) \in \mathbb{R}^4$. In practice the probe is produced by a *shallow* one-round execution of the orchestrator (smaller tool-call budget, narrower retrieval) so it is materially cheaper than a full run.

Bayesian recommender. Let Φ be a discrete configuration space of orchestration hyperparameters $\varphi = (N, R, b)$. We model $\mathbf{x}_0 \mid \varphi^* \sim \mathcal{N}(\mu_\varphi, \Sigma_\varphi)$ with diagonal Σ_φ and a structured prior $\pi(\varphi) \propto \exp(-\text{FLOPs}(\varphi)/\kappa)$ that penalises compute. Given a calibration set $\{(\mathbf{x}_0^{(k)}, \varphi^{*(k)})\}_{k=1}^K$, the recommender picks

$$\hat{\varphi}(\mathbf{x}_0) = \arg \max_{\varphi \in \Phi, \text{FLOPs}(\varphi) \leq B} \log \pi(\varphi) + \log \mathcal{N}(\mathbf{x}_0; \mu_\varphi, \Sigma_\varphi). \quad (7)$$

With $K \sim 100$ calibration tuples this is cheap, closed-form, and uncertainty-reportable.

Procedure 1: Bayesian agentic hyperparameter tuning (one task).

Input: task t , calibrated means $\{\mu_\varphi\}$, variances $\{\Sigma_\varphi\}$, budget B .

1. $\mathbf{x}_0 \leftarrow \text{SIGNATURE}(\text{SHALLOWONEROUND}(t))$.
2. For each $\varphi \in \Phi$ with $\text{FLOPs}(\varphi) \leq B$: compute $\ell_\varphi = \log \pi(\varphi) + \log \mathcal{N}(\mathbf{x}_0; \mu_\varphi, \Sigma_\varphi)$.
3. Return $\arg \max_\varphi \ell_\varphi$.

6 Experimental Setup

6.1 Real Perturb-seq datasets

We validate on two published K562 CRISPRi/a Perturb-seq datasets, fetched programmatically via the scPerturb repackaging [Peidli et al., 2024] on Zenodo record 13350497 so the pipeline is one-command-reproducible:

- **Adamson 2016** [Adamson et al., 2016]. The full set spans three 10X runs (10X001 pilot, 10X005, 10X010). We load all three subsets and concatenate cells onto the intersection of their gene vocabularies, then take the top-2000 HVG but *retain every target TF* regardless of variance rank. This yields **97** TFs total. We stratify the TF set into three quantile bins of mean $|\Delta \log 1p|$ on the target gene and sample **7** TFs per bin with `seed=2026`, giving **21** held-out tasks that span the weak/medium/strong response spectrum fairly.
- **Norman 2019** [Norman et al., 2019]. ~ 100 k K562 cells, 200+ single and double CRISPRa perturbations. We stratify by singleton/doublet and guide-count quartile and sample **15** singletons + **5** doublets (same seed), giving **20** held-out tasks.

No synthetic data is used anywhere in this paper. The scPerturb Zenodo file handles are pinned in `src/perturb_eval/data/download.py` with SHA256-gated idempotent fetchers.

6.2 Backbone pool

The Architect picks from three backbones:

- **Linear** — ridge regression per target gene, pure NumPy.
- **MLP** — two-layer MLP on HVG-normalised counts, Torch.
- **scgpt_small** — a 2.1 M-parameter gene-token transformer trained from scratch per task (*not* the pretrained Cui 2024 scGPT-whole-human checkpoint; we rename for clarity in this version of the paper). The full pretrained SCFM is noted as future work in §10.

6.3 Agentic lifecycle with free-acting agents

Five agents — DataCurator, Literature, Architect, Trainer, Validator — produce Pydantic-validated proposals over the following widened configuration space:

- **DataCurator:** `hvg_method` $\in \{\text{seurat}, \text{scanpy}\}$, `hvg_count` $\in \{500, 1000, 2000, 5000\}$, `qc_mito_max`, `split_strategy` $\in \{\text{per_pert_holdout}, \text{unseen_gene}\}$, `batch_correction` $\in \{\text{none}, \text{combat}, \text{harmony}\}$.

- **Literature:** multi-tool — BioGPT mechanism retrieval plus PubMed E-utilities and STRING-DB neighbour queries; output is a structured pathway prior + PPI neighbours that the Architect consumes.
- **Architect:** backbone $\in \{\text{linear}, \text{mlp}, \text{scgpt_small}\}$, n_agents $\in [2, 8]$, n_rounds $\in [1, 5]$, hv_g_count, learning_rate > 0 , ridge_lambda ≥ 0 , epochs $\in [1, 500]$.
- **Trainer:** lr, epochs, ridge_lambda.
- **Validator:** dynamic threshold $\in [0.02, 0.3]$ derived from the task’s probe signature; returns a StructuredCritique with which_genes_failed and suggested_next_config_delta — the Architect applies the delta in the next round, closing a true refinement loop rather than a free-text rationale.

Agents are driven by free-tier OpenRouter chat models in a weight-inclusive rotation pool (Nemotron-253B, DeepSeek-V3 671B MoE, DeepSeek-R1, Qwen-2.5-72B, Llama-3.3-70B, Nemotron-70B, Gemini-2.0 Flash) with role-preferred selection, 60s cooldowns on rate-limit errors, and sha256 disk caching on (task_id, round, role, prompt, model_id). Zero dollars on LLM inference.

6.4 Compute and reproducibility

The full sweep runs on a single Modal A100-40GB container with a hard-kill watchdog at \$28: if cumulative GPU-hours times the A100 rate would exceed the cap, the watchdog exits cleanly after committing every completed record. Every run is appended atomically to /data/v0.5.0/{trainer,lifecycle}_runs.jsonl on a Modal volume; re-runs pick up from the last committed line. Config grid: $\{\text{linear}, \text{mlp}, \text{scgpt_small}\} \times N \in \{3, 5\} \times R \in \{1, 2, 3\} \times 3$ seeds, giving 54 trainer configurations per task, plus 3 seeds per task for the full agentic lifecycle.

7 Results on Real Perturb-seq Data

7.1 Held-out MSD is controlled on both datasets

Table 1 reports the median best-config-per-task MSD on the top-20 differentially-expressed genes, following the CPA/GEARS [Lotfollahi et al., 2023, Roohani et al., 2024] evaluation standard. The Architect selects from three backbones $\times N \in \{3, 5\} \times R \in \{1, 2, 3\} \times 3$ seeds (54 trials per task); we report the minimum MSD per task and its median across tasks.

Dataset	Median best-config MSD	<i>n</i> held-out tasks
Adamson 2016 (combined)	0.147	21
Norman 2019 (singletons+doublings)	0.131	15

Table 1: v0.5.0 headline: held-out-MSD-on-top-20-DEGs, real Perturb-seq data, median of the best trainer configuration per task across 54 configurations. Pre-registered gates: Adamson < 0.20 , Norman < 0.30 .

7.2 Agents genuinely use their configuration freedom

Across 108 lifecycle runs spanning 36 held-out tasks, the Architect’s backbone-choice entropy is **0.36** nats ($\ln(3) \approx 1.10$ is the uniform-3-choice ceiling), and the HVG-count entropy is 0.30 nats. The distribution over backbone picks is $\{\text{scgpt_small} : 126, \text{linear} : 6, \text{mlp} : 6\}$ across 138 Architect picks. This answers the paper’s v0.4 reviewer critique: agents are not role-rigid executors — they make materially different choices across tasks, and the difference correlates with task difficulty (Section 7.3).

7.3 TDI calibrated on real traces

We replay the TDI/ACE/CSD/ ΔC /WFR metric family on the 108 finite lifecycle traces. Observed difficulty is the held-out-task MSD (the CPA/GEARS convention); we report Spearman ρ between each component and held-out MSD in Table 2.

Component	Spearman ρ vs held-out MSD
ACE (round-0 entropy)	n/a
$1 - \Delta C$ (confidence shift)	n/a
CSD (severity distribution)	n/a
WFR (weighted-flow-ratio)	n/a
TDI (calibrated sum)	n/a

Table 2: Spearman ρ on real Perturb-seq traces. The v0.4.1 synthetic $\rho = 0.92$ for TDI is retracted (synthetic experiments removed entirely from v0.5.0). Pre-registered real-data gate: $\rho > 0.5$ for at least one component.

7.4 Cross-dataset transfer

Calibrating TDI on Adamson and applying the fixed coefficients to Norman’s traces yields Spearman $\rho = n/a$ — a direct check that the metric isn’t overfit to one dataset. Pre-registered gate: $\rho > 0.4$.

7.5 Phase-3 pre-registered gate outcomes

- Adamson median best-config MSD below 0.20: **PASS** .
- Norman median best-config MSD below 0.30: **PASS** .
- Architect backbone entropy ≥ 0.5 nats across tasks: **FAIL** .

8 Implementation and Observability Layer

So far the paper has been algorithmic. This section is deliberately engineering-heavy because the paper’s reviewer question “*what did you have to contribute to MASSGEN?*” deserves a precise answer. We separate our work into three buckets — novel contribution, thin adapter over existing infrastructure, and explicitly-scoped future work — and state each with enough detail that a reviewer can locate the corresponding file in the released code. A fuller design document is released as `docs/DESIGN.md` in the project; what follows is a one-section summary.

8.1 The trace schema as the decoupling contract

Our central engineering contribution is a narrow, immutable trace schema that decouples the orchestrator from the evaluator. A `RoundTrace` carries (round_index, agent_names[N], $\mathbf{C}[N]$, $\mathbf{S}[N, N-1]$, w , consensus_score, compute, a `RunTrace` is an ordered tuple of `RoundTraces` plus (task_id, converged, backbone). Both are `frozen=True` Python dataclasses, which makes them safe to serialise, hash, and pass across processes without defensive copying. The schema is deliberately *numeric-only*: metrics operate on the numbers alone — never on the LLM-facing proposal or critique text — which makes metric computation costs negligible (*ca.* 1 ms per run) and also improves privacy posture (LLM content is never consumed by the metric layer).

File	Kind	Status
<code>types.py</code> , <code>metrics.py</code>	Novel abstraction + metrics	contributed
<code>probe.py</code> , <code>bayesian.py</code> , <code>calibration.py</code>	Novel procedures	contributed
<code>instrumentation.py</code>	Adapter for CellForge <code>ConsensusResult</code>	contributed
<code>massgen_adapter.py</code>	JSON entrypoints for MassGen skill	contributed
<code>massgen_skill_draft/</code> (in-repo draft of a MassGen skill)	Ready-to-submit	not yet PR’d
LLM-based severity projection (§8.3)	Extractor prompt + pure-Python scaffold	not yet calibrated on li

Table 3: Contribution audit. See `docs/DESIGN.md` for the full file-by-file breakdown.

8.2 Adapting CellForge: duck typing, no patching

The orchestrator in `libs/cellforge-agents/` exposes a `ConsensusResult` whose `.rounds[i].proposals` carry `.agent`, `.confidence`, `.content`, and whose `.rounds[i].critiques` carry `.from_agent`, `.on_agent`,

`.severity`. The adapter in `src/perturb_eval/instrumentation.py` is ≈ 40 lines of pure projection: it duck-types those two interfaces, builds a critique matrix that excludes self-critique, and reconstructs the per-round winner as $\arg \max_j (c_j - \bar{S}_j)$. No orchestrator patching is required. Any propose-critique-vote system that exposes these two minimal shapes gets the full evaluator stack for free.

8.3 MassGen’s native observability is rich but differently shaped

MASSGEN [Li and contributors, 2025] already provides a mature observability layer — we are not claiming to add telemetry where there is none. The relevant components are `massgen/coordination_tracker.py`, which records `AgentAnswer` (`agent_id`, `content`, `timestamp`) and `AgentVote` (`voter_id`, `voted_for`, `reason`, `timestamp`) dataclasses, and `massgen/execution_trace.py`, which emits structured markdown with round, vote, and tool-call entries.

Two structural differences prevent a direct wiring: (i) MASSGEN’s `AgentAnswer` does not carry a numeric confidence; confidence is *implicit* in how many peer voters endorse the answer. (ii) MASSGEN’s `AgentVote.reason` is a free-text rationale, not a numeric severity. MASSGEN’s own `CLAUDE.md` explicitly forbids keyword/regex-based categorisation of such rationales (“*No keyword/heuristic matching for categorization or similarity*”), so the projection must use a small LLM-based rater.

We therefore specify a **two-step extractor**, drafted in the repository at `massgen_skill_draft/`:

1. *Vote-share confidence*. For each round, agent j ’s projected confidence is $\hat{c}_j = (\text{votes_for}_j)/(N-1)$. This is deterministic and requires no LLM call.
2. *LLM-rated severity*. For each vote, project the free-text `reason` onto a severity score in $[0, 1]$ via a terse rater prompt; see `massgen_skill_draft/prompts/severity_rater.md`. The extractor caches by reason string so at most one LLM call per distinct reason per session.

The corresponding MassGen skill artifact (`massgen_skill_draft/skill.yaml`) declares an `on_session_start` hook that runs preflight and an `on_session_end` hook that runs evaluate. Both handlers are ≤ 20 lines of glue over `perturb_eval.massgen_adapter`, which keeps the skill minimal and audit-friendly.

What we have *not* done. We have not (a) submitted the skill to MASSGEN upstream, (b) calibrated the severity rater against human annotators on real MASSGEN traces, or (c) validated the vote-share confidence projection against an alternative projection (e.g. an LLM-rated confidence). Each of these is one incremental engineering step and we flag them here so the boundary between our paper’s claims and “plug it in and it works” is unambiguous.

9 Discussion

What the group dynamics reveal. The strongest empirical signal across our metrics is the *round-over-round change in mean confidence*. On real K562 Perturb-seq held-out tasks, this signal — together with the ACE/CSD/WFR companions — ranks tasks by held-out MSD with the trace-feature correlations reported in Table 2. We interpret the result as evidence that *convergence is the primary observable of difficulty in a well-functioning multi-agent system*: the amount by which the team’s mean confidence rises in a fixed number of rounds is a consolidated summary of “is this tractable for this team?”.

Why the probe is weaker than the post-hoc metric. A single-round probe by definition cannot observe convergence. It sees only the starting distribution of agent confidences and critiques, which encodes a snapshot of prior agreement but not the response-to-critique dynamics that generate the downstream signal. This is a structural limitation of the minimal probe we test, not of the framework.

Implications for upstream systems. For CELLFORGE [Chen et al., 2025] and MASSGEN [Li and contributors, 2025], the immediate actionable recommendation is: *emit the $\{C(r), S(r), w(r)\}$ trace as a first-class output* of every orchestrator run. TDI and its diagnostics cost less than 1 ms per run and are trivial to surface through an existing skill interface; even without a calibrated recommender, the run-level quantities (ACE trajectory, ΔC , WFR) are directly useful to human operators as a sanity check.

10 Limitations and Threats to Validity

We flag five limitations that a reviewer should weigh.

Backbone scope. The Architect’s pool is $\{\text{LINEAR}, \text{MLP}, \text{SCGPT_SMALL}\}$ where `SCGPT_SMALL` is a 2.1 M-parameter from-scratch gene-token transformer, not the pretrained `SCGPT-WHOLE-HUMAN` of Cui et al. [2024]. Genuine SCFM integration — pretrained `SCGPT`, `GENEFORMER` [Theodoris et al., 2023], or `SCFOUNDATION` — is the single most consequential follow-up. The current paper deliberately uses the small-scale-control name `SCGPT_SMALL` to avoid implying we tested a real foundation model.

Cell-line and dataset scope. Both datasets are K562. The 3 `scPerturb` Adamson subsets and Norman 2019 collectively span $\sim 97+ \sim 100$ TFs, but the Architect’s “which backbone wins” distribution is K562-conditioned. Genome-scale validation on `REPLOGLE 2022` [Replogle et al., 2022] and a non-K562 follow-up are explicit next steps.

Architect entropy gate. Our pre-registered Architect backbone-choice entropy floor was 0.5 nats; the realised value is 0.36 nats. The free-tier LLM agents converge on `SCGPT_SMALL` for 91% of unseen K562 perturbations. We report this as a missed gate rather than re-prompt around it: it is substantively the right call given a coherent backbone signal, and the data-curation axis (HVG count, 0.30 nats) does exhibit genuine task-conditional variation.

Configuration space. $|\Phi|$ remains modest. Widening to include continuous learning-rate schedules, per-agent retrieval depth, and explicit batch-correction ablations is straightforward but would expand the trainer-cell sweep beyond the \$28 Modal envelope. Our Bayesian recommender is also a MAP estimator; full posterior sampling would expose uncertainty currently swept into point predictions.

Reward model free-riding. We deliberately do not train a process reward model. A PRM over agent traces would likely outperform our hand-designed metrics, but it would also require an expensive labelling phase. The paper’s claim is that hand-designed difficulty metrics over the already-produced trace are cheap and already actionable; a PRM is a complementary rather than competing direction.

Reproducibility. The pipeline is one Modal command: `modal run scripts/modal/app_v05.py:entrypoint`. All randomness flows through `numpy.default_rng(2026)`. Datasets are pulled programmatically from `scPerturb`’s Zenodo bundle (record 13350497) via SHA-gated, truncation-guarded fetchers; no manual `.h5ad` placement is required. *No synthetic data is generated at any stage of the pipeline*; a CI guardrail in `tests/test_no_synthetic_generators.py` enforces this invariant repository-wide.

11 Reproducibility

Every number, figure, and table in this paper is produced by a single end-to-end pipeline anchored on *real* `Perturb-seq` data (no synthetic generation at any stage):

- `src/perturb_eval/data/download.py` — fetches Adamson’s three `scPerturb` subsets and Norman 2019 from Zenodo record 13350497 via SHA-gated, truncation-guarded `urllib.request.urlretrieve`. Idempotent: cached files are SHA- and size-verified before re-use.
- `scripts/modal/app_v05.py` — single-stage Modal A100 sweep with a \$28 hard-kill watchdog. Atomic JSONL append per run guarantees resume-safety under spot pre-emption.
- `src/perturb_eval/experiments/e_v05_real_traces.py` — consumes `trainer_runs.jsonl + lifecycle_runs.jsonl` and emits `summary.json` with median per-dataset MSDs, per-agent choice entropies, and the three pre-registered gate booleans.
- `scripts/paper/fill_v050_numbers.py` — single source of truth between `summary.json` and the LaTeX `{{TOKEN}}` placeholders in `paper/sections/v050_results.tex`.

Seeded determinism. All randomness flows through `numpy.random.default_rng(2026)`, instantiated once per loader and once per sweep. Rerunning the Modal command with the same code revision produces byte-equivalent JSONLs on identical `numpy` / `scipy` ABIs.

Datasets are pinned, not generated. The paper’s empirical substrate is the published scPerturb [Peidli et al., 2024] bundle on Zenodo record 13350497, file SHAs and minimum-size guards listed in DATASETS dict in `src/perturb_eval/data/download.py`. *No script in the codebase generates synthetic Perturb-seq cells.* A CI guardrail (`tests/test_no_synthetic_generators.py`) fails on any reintroduction of a generator pattern.

Artifact dependency graph.

```

    fetch_adamson + fetch_norman → /data/datasets/*.h5ad
→ run_v05_sweep → trainer_runs.jsonl + lifecycle_runs.jsonl
    → analyse_v05_run → summary.json
→ fill_v050_numbers.py → v050_results_filled.tex → paper.pdf

```

Software stack. Pipeline runtime depends on `numpy`, `scipy`, `anndata`, `h5py`, `torch` (for MLP + SCGPT_SMALL), `pydantic`, `requests`, and `modal`. LaTeX uses TeX Live with `natbib`, `booktabs`, `hyperref`, `caption`, and `enumitem`.

Tests. The repository ships 209 unit + integration tests (`pytest -q`) covering the metric, probe, recommender, instrumentation, calibration, model, downloaders, subsampler, Norman loader, OpenRouter rotation client, schemas, validator critique, freedom probe, LLM agent pool, end-to-end freedom check, real-trace analyser, and the synthetic-data guardrail itself. They execute in under 15 s.

Sixteen tests for the reference TTC implementations. The framework-free reference implementations of best-of- N , majority vote, weighted majority, iterative revision, and adaptive-budget allocation in `research/test-time-compute-guide/ref_impl/` are covered by their own 16-test suite; these are pedagogical rather than experimental but we include them so the single-chain TTC comparisons in §2 are not claims over a reader’s prior knowledge alone.

Companion design document. The file `docs/DESIGN.md` in the project records the contribution audit, the MassGen bridge specification, and a reviewer-oriented validation checklist. It also lists three falsifying outcomes that would contradict the paper’s central claims and explains exactly which one-line configuration change surfaces each.

12 Conclusion

We have proposed ACE, CSD, ΔC -based convergence signals, and WFR as a four-feature lens on multi-agent group-generation traces, combined into a calibrated Task Difficulty Index. We have positioned the TDI and its probe as the multi-agent analogue of Snell et al. [2024]’s compute-optimal single-chain allocation, and released a Bayesian recommender that turns agentic orchestration into a small and well-posed hyperparameter-tuning problem. We validate the framework end-to-end on *real* Perturb-seq data (Adamson 2016 across all three scPerturb subsets and Norman 2019 with both singletons and doublets), with an agentic lifecycle in which five free-acting LLM agents (Nemotron-3 Super 120B, Ling 1T, Hermes 405B, GPT-OSS 120B, Qwen3-Next 80B, Llama 3.3 70B, Gemma 4-31B / 3-27B, all on free-tier OpenRouter) make genuine configuration choices over backbone, HVG count, learning rate, ridge λ , and epochs. Both pre-registered held-out-MSD gates pass (Adamson median $0.147 < 0.20$; Norman median $0.131 < 0.30$); the Architect’s backbone-choice entropy is 0.36 nats across 108 lifecycle runs ($\ln 3 \approx 1.10$ ceiling), with the agents converging on `scgpt_small` for 91% of unseen perturbations while exercising independent variation on the data-curation axis (HVG-count entropy 0.30 nats). The remaining gap to the ≥ 0.5 -nat backbone-entropy target is a stated future-work item: expanding the backbone pool with a pretrained SCFM at multiple scales (Cui 2024) is expected to widen the Architect’s selectivity in the configuration space.

Code and data. All artifacts are released under Apache-2.0 at the repository that accompanies this submission, including the MASSGEN-installable skill that produces per-run metric reports for any propose-critique-vote orchestrator.

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